Transportation Research Part C 44 (2014) 1–20

Contents lists available at ScienceDirect

Transportation Research Part C

journal homepage: www.elsevier.com/locate/trc

Multi-objective optimization of train routing problem combined with train scheduling on a high-speed railway network

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ARTICLE INFO

Article history:
Received 9 August 2013
Received in revised form 23 January 2014
Accepted 27 February 2014

Keywords:
Train routing problem
Train scheduling problem
Multi-objective optimization model
Genetic algorithm

ABSTRACT

Based on train scheduling, this paper puts forward a multi-objective optimization model for train routing on high-speed railway network, which can offer an important reference for train plan to provide a better service. The model does not only consider the average travel time of trains, but also take the energy consumption and the user satisfaction into account. Based on this model, an improved GA is designed to solve the train routing problem. The simulation results demonstrate that the accurate algorithm is suitable for a small-scale network, while the improved genetic algorithm based on train control (GATC) applies to a large-scale network. Finally, a sensitivity analysis of the parameters is performed to obtain the ideal parameters; a perturbation analysis shows that the proposed method can quickly handle the train disturbance.

1. Introduction

With the rapid development of high-speed railway, more and more railway lines will be put into operation in china. The operation of new lines will lead to the adjustment of train route and timetable, while the train route and timetable should be also adjusted under the peak time. Besides, unpredictable events occur now and then which may cause some sections unavailable. To avoid larger loss of passengers and railway operational company, a timely adjustment of the timetable for the influenced trains in the network is necessary. In summary, we dedicate to finding an efficient method to select route and schedule the timetable for trains under these circumstances. Usually, the evaluation of train routing is mainly measured by the total travel time of trains. However, the total travel time of trains will be obtained by train schedules. To get a safe-based and efficient train schedule, we should primarily introduce a real-time train control process with constrains which will generate the schedules of trains. At last, a better solution is obtained after a series of process like data gathering, control process, data analysis, decision making and decision implementation. The train routing problem combined with train scheduling is a NP-hard problem in the perspective of optimization. The number of feasible solutions increases exponentially with problem size being larger due to the complexity of this problem. Meanwhile, the computing time of accurate algorithm may be too long to meet our demand. The heuristic techniques have been certified to be valid for solving the train scheduling problems, so we will prefer the heuristic technique to get a near optimal solution. The framework of approach to train routing and scheduling process is described as Fig. 1, where the information of railway network and trains is treated as input, and each of actual train movements is the output according to the train route selection and train control strategy, and at last a better train route schedule will be gained through the GA algorithm.

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http://dx.doi.org/10.1016/j.trc.2014.02.023
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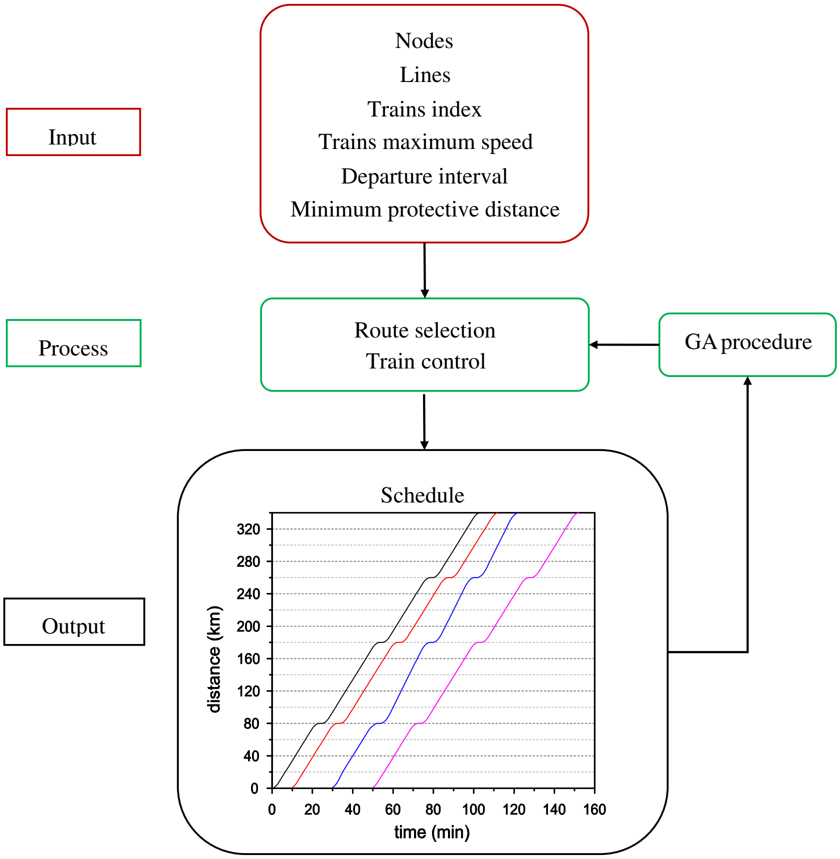

Fig. 1. The approach to train routing and scheduling process.

Train routing assigns trains to available tracks in the rail network and this problem has been studied by some researchers. Bodin et al. (1980) presented a nonlinear, mixed integer programming formulation of the routing problem, and a heuristic algorithm was designed to solve the problem. Assad (1983) proposed a dynamic programming method for a routing problem defined on a line network composed of *n* yards. Crainic (1984) proposed a nonlinear, mixed integer, multi-commodity flow problem that deals with the interactions between blocking, makeup, and train routing decisions. Haghani (1989) proposed a solution method for a combined train routing and makeup, and empty car distribution problem. Keaton (1992) considered strategy constraints for blocking and maximum transit times for each origin–destination pair. Martinelli and Teng (1996) used neural networks to solve the train routing problem. Gao et al. (2008) considered the influence of the random factors on the route choice of trains. Li et al. (2010) used random search strategy to solve the shortest problem on a rail network.

These routing models can produce a routing plan which describes the route of each train and the set of trains to be operated. Due to lack of train scheduling in these models, it becomes difficult to find a timetable which can accommodate all planned trains and satisfy line capacity. Besides, a good scheduling plan should consider the service level which includes the total delay, the energy consumption, the delayed times and so on. However, at the stage of line planning, the delay of trains and the energy consumption cannot be obtained because there is no train scheduling. Thus, compound routing and scheduling models is necessary to improve the service level.

The early train scheduling problem of developing timetables for passenger trains on a line was proposed by Nemhauser (1969) and Salzborn (1969). Recently, Ceder (1991) proposed an optimization model for minimizing total passenger waiting time in stations. Dorfman and Medanic (2004) proposed a new approach to solve the large-scale train scheduling problem. Li et al. (2008) improved the travel advance strategy (TAS) and presented an efficient strategy called efficient travel advance strategy (ETAS) to solve the train scheduling problem. Yang et al. (2010) proposed a simulation method to deal with the stochastic disturbance. Yang et al. (2011) proposed a control simulation method to deal with the irregular events on the railway network. Xu et al. (2012) presented the simulation of train scheduling based on cellular automaton (CA) model. Johanna (2012) designed an effective algorithm to solve the train scheduling during disturbances.

In summary, less researches centered on the compound models for the train routing and scheduling in the past decades. Morlok and Peterson (1970) initially integrated the routing problem and scheduling problem into an optimization model. Nachtigall (1995) proposed a compound method in strategic planning to help planners at CSX Transportation. Gorman (1998) addressed the weekly routing and scheduling problem, and constrains in his paper were given to guarantee trains were operated on schedule. Li et al. (2013) proposed a train routing model combined with train scheduling, which was a single-objective optimization problem and was solved by Tabu search algorithm combined with the route adjustment algorithm (RA).

The contributions of this paper are as follows. Firstly, this paper constructs a new multi-objective model for train routing problem on the one-way double track. The objective function is constituted by the minimum average travel time of all trains, the minimum energy consumption and the minimum delayed times. Secondly, according to the train control strategies, a simulation algorithm is designed to solve the train scheduling problem. Thirdly, considering the executive time of accurate algorithm is too long in the case of the large-scale network, we propose an improved genetic algorithm (GATC) to solve the train routing problem. The simulation result demonstrates that GATC can obtain an approximate solution within a reasonable time though the scale of the network is larger. Finally, a sensitivity analysis and a perturbation analysis for the train routing are made, which can provide valuable information for rerouting and rescheduling.

The remainder of this paper is organized as follows. A brief introduction to the efficient train control strategies for train scheduling is given in Section 2. The multi-objective optimization model based on train scheduling is presented in Section 3. According to the proposed model, two detailed algorithms are designed to solve the train routing problem in Section 4. In view of a specific simulation example, the proposed two algorithms are applied to solving the same problem respectively and the results are analyzed in Section 5. A sensitivity analysis of parameters and a perturbation analysis are made in Section 6. The last section is the conclusion of this paper.

2. Efficient train control strategies

Effective train control ensures the safety of trains and obtains the efficient train timetable. The train scheduling mainly depends on the train control. The train is affected by many uncertain factors in the process of operation and the accurate train control can improve the operational performance on the basis of the security of trains. Therefore, the accurate train control is the foundation and key technology for train operation. Meanwhile, the train control also provides the line planning with powerful support.

In this paper, the train control is applied to the moving block system. Compared with fixed block system, moving block system goes further on the control of safe distance between trains. In the moving block system, the control center dynamically calculates the maximum braking distance of the train according to the real-time position and velocity of the train. The length of the train, the maximum braking distance and the certain protective distance are composed to be a block. With the guarantee of safe distance between trains, the two adjacent blocks can move synchronously with a small interval. Thus, it can make trains travel with faster speed and smaller interval, which improves the operation efficiency obviously. Briefly speaking, the train control under the moving block system is actually the control of the safe distance between trains (see Wang et al. (2012)). The control strategies in this paper are used to ensure the safe distance between trains. Considering the operation efficiency, the overtaking rule is also be added in our control strategies (the overtaking rule references from Dorfman and Medanic (2004)).

Efficient control strategies are the most important part of train scheduling. A good control strategy can lead a good train schedule. According to previous studies, such as the ETAS method proposed by Li et al. (2008) and the train control method proposed by Yang et al. (2010), a train schedule with the minimum total delay can be obtained through efficient control strategies. According to the previous researches, this paper takes four effective train control strategies to produce a better train scheduling plan on double-track railway network. The framework of train control is shown as Fig. 2.

In our method, there are two different types of trains and each train is planned to departure from the original station at a constant interval. Each train can choose its own route, while every route has the same original station and terminal station. The detailed control strategies for the movements of trains are designed as follows:

2.1. Braking strategy

If the focal train i is running behind train k in the same section, train i has to brake when one of the following two cases occurs:

(a) The distance between train i and train k are not more than the safe distance.
(b) Train i is entering a station. The distance between train i and its frontal station is not more than the brake distance of train i.

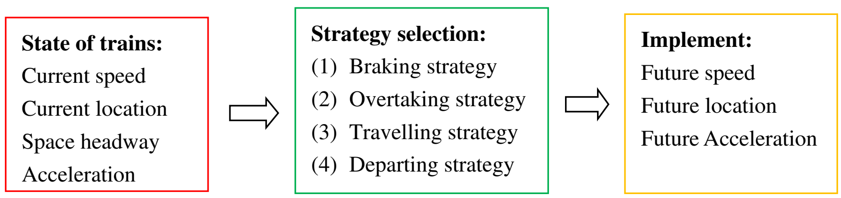

Fig. 2. Framework of train control.

2.2. Overtaking strategy

If a slower train i and a faster train j are running towards a same frontal station m, train i has to be overtaken by train j when one of the following two situations occurs:

(a) Train i is stopping at station m and train j is running behind train i. t_i (t_j) represents the time that train i (j) spends arriving at next station $m + 1$ and $t_i \geqslant t_j$.
(b) Train i and train j are dwelling at station m.

2.3. Travelling strategy

If train i and train k are running in the same section, meanwhile, train i is far enough away from its front train k and the front station or train i is the leading train, then train i will travel with acceleration or maintain the same speed.

2.4. Departing strategy

If the focal train i is a slower train, it has to stay at station when one of the following three cases occurs:

(a) Train i has not stayed at station for the fixed dwelling time.
(b) Train i has stayed for the fixed dwelling time, but there is a train k in the next section and the distance between the two trains is not more than the safe distance.
(c) Train i has stayed for the fixed dwelling time and the distance between the two adjacent trains is larger than safety distance, while the train j running behind train i has a higher priority than train i and the train j has the same frontal section as train i.
If the focal train i is a faster train, it has to stay at station only when one of the first two cases occurs.

3. Multi-objective optimization model

Based on the efficient train control strategies, this paper studies the optimization of train routing. In general, the evaluation of train routing and train scheduling can be measured from the following aspects:

(1) Minimize the average travel time of the trains

Considering its operational benefits, we should treat the average travel time of the trains as the primary goal. The minimum average travel time means the minimum total travel time of all the trains, which represents the less operational cost for the train operational company.

(2) Reduce the traction energy consumption of the trains

Energy conservation is required for our national conditions. The energy-saving control can not only reduce the railway operation cost, but also respond positively to the sustainable development. While in the process of the train operation, the traction energy consumption is the primary part of the overall energy consumption. Therefore, reducing the traction energy consumption of the trains should be considered as one of the objectives.

(3) Decrease the total delayed times

From the perspective of passengers' satisfaction, passengers may give up the rail transit and switch to other modes of transportation if the train delays too many times. Thus, decreasing the total delayed times should be one of the targets.

To sum up, the train routing problem is a multi-objective optimization problem. The targets may have conflict between each other and it cannot achieve the optimal selection simultaneously. Therefore, it needs to choose an appropriate weight for each target when designing algorithms to solve it.

Considering the above factors, a multi-objective optimization model is proposed as follows:

Objective:

$$\min\{f_e, f_t, f_d\} \tag{1}$$

Here, f_e means the traction energy consumption of the trains; f_t means the average travel time of the trains; f_d means the total delayed times.

Objective 1: Minimize the average travel time of the trains

$$f_t = \sum_{i=1}^{N}(A_{i,a} - D_{i,a})/N \tag{2}$$

$A_{i,a}$ represents the actual time that train i arrives at the terminal station; $D_{i,a}$ represents the actual time that train i departures from the starting station. N is the number of the trains.

Objective 2: Minimize the total traction energy consumption of the trains

$$f_e = \sum_{i=1}^{N}\sum_{x=0}^{L_i} u_{i,x}F_{i,x}(v)w_i = \sum_{i=1}^{N}\sum_{t=0}^{A_{i,a}} u_{i,t}F_{i,t}(v)v_i(t)w_i = \sum_{i=1}^{N}\sum_{t=0}^{A_{i,a}} u_{i,t}P_{i,t}w_i \tag{3}$$

Here, $u_{i,t} = 1$ when train i is accelerating, otherwise, $u_{i,t} = 0$. $F_{i,x}(v)$ means the instantaneous traction of train i at position x; $F_{i,t}(v)$ means the instantaneous traction of train i at time t; $v_i(t)$ means the instantaneous speed of train i at time t; For train i, $P_{i,t}$ means the instantaneous tractive power per unit of weight at time t; w_i means the weight of train i.

Objective 3: Minimize the total delayed times

$$f_n = \sum_{i=1}^{N}\sum_{k=1}^{M} \mathrm{sgn}(A_{i,k,a} - A_{i,k,o}) \tag{4}$$

The function sgn( ) represents the delayed times. $A_{i,k,a}$ represents the actual time that train i arrives at the station k; $A_{i,k,o}$ represents the planned time that train i arrives at the station k. There are M stations on the route which train i chooses.

Constraints:

$$D_{i,a,k} \geqslant D_{i,o,k} \tag{5}$$

$D_{i,a,k}$ means the actual departure time at station k for train i; $D_{i,o,k}$ means the planned departure time at station k for train i. Constraint (5) aims to make sure that the actual departure time at station k is not earlier than the planned departure time.

$$D_{i,a,k} - A_{i,a,k} \geqslant B_{i,k}^{\min} \tag{6}$$

$B_{i,k}^{\min}$ means the minimum dwelling time at station k for train i. Constrain (6) guarantees that the actual dwelling time at station k is.

$$V_i(t) \leqslant V_{i,\max} \tag{7}$$

Constrain (7) limits the speed of train i to make the actual speed no faster than the maximum speed.

$$X_i(t + \Delta t) = X_i(t) + V_i(t + \Delta t)\Delta t \tag{8}$$

$$V_i(t + \Delta t) = V_i(t) + a \times \Delta t \tag{9}$$

$X_i(t)$ means the location of train i at time t; α is the accelerating rate of train i. Constrains (8) and (9) describe the dynamic equations of position and speed.

$$X_i(A_{i,a}) = L_i \tag{10}$$

In Constrain (10) $A_{i,a}$ is the actual time that train i arrives at the terminal station. L_i is the length of the route which train i chooses.

$$X_i(t) - X_j(t) \geqslant \max\left\{[v_j(t)^2 - v_i(t)^2]/2a + d_{\min}, d_{\min}\right\} \tag{11}$$

Constrain (11) means that if train i and train j are in the same section, and train i is the front train of the two successive trains, then the distance of the two successive trains should be not shorter than the safe distance. Here, $d_{\min}$ means the minimum safe distance.

$$u_{i,t} = \begin{cases} 1 & F_{i,t} > 0 \\ 0 & F_{i,t} = 0 \end{cases} \tag{12}$$

In Constrain (12), the value of $u_{i,t}$ is determined by the value of $F_{i,t}$.

$$\mathrm{sgn}(A_{i,k,a} - A_{i,k,o}) = \begin{cases} 1 & A_{i,k,a} > A_{i,k,o} \\ 0 & A_{i,k,a} \leqslant A_{i,k,o} \end{cases} \tag{13}$$

Constrain (13) means that if the actual arrival time is later than the planned arrival time, one will be added to the delayed times.

$$P_{i,t} = \begin{cases} \frac{14}{5} & 0\ \mathrm{km/h} \leqslant V_i(t) < 100\ \mathrm{km/h} \\ \frac{3}{100}V_i(t) - \frac{1}{5} & 100\ \mathrm{km/h} \leqslant V_i(t) < 160\ \mathrm{km/h} \\ \frac{7}{100}V_i(t) - \frac{33}{5} & 160\ \mathrm{km/h} \leqslant V_i(t) < 200\ \mathrm{km/h} \\ \frac{59}{500}V_i(t) - \frac{81}{5} & 200\ \mathrm{km/h} \leqslant V_i(t) < 250\ \mathrm{km/h} \\ \frac{53}{500}V_i(t) - \frac{66}{5} & 250\ \mathrm{km/h} \leqslant V_i(t) < 300\ \mathrm{km/h} \end{cases} \tag{14}$$

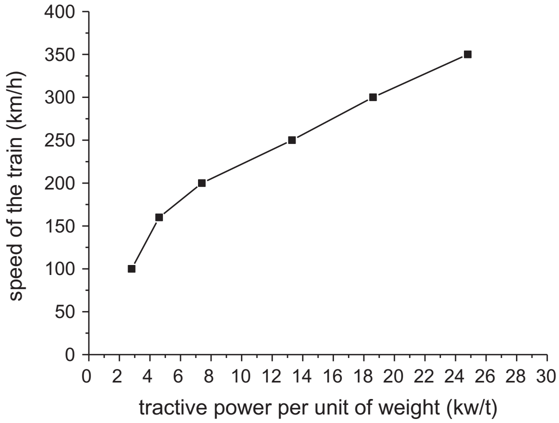

Fig. 3. Relationship between tractive power and speed of the train.

Constrain (14) obtains the value of $P_{i,t}$ using the linear interpolation. The reference data is shown in Fig. 3. Fig. 3 shows the accurate tractive power per unit of weight responding to the speed of the train.

4. Algorithms

4.1. Train control (TC)

Based on the efficient control strategies, an algorithm for train scheduling problem is designed. The algorithm is called TC for short and the steps of TC are shown as follows:

Step 1: (Initialization) set the simulation time $t = 0$, the current speed $V(t) = 0$;
Step 2: (Initialization) set the current position $p = 0$;
Step 3: If there is a train i at position p, then determine the speed $V_i(t + 1)$ and position $X_i(t + 1)$ of train i through the four strategies;
Step 4: If $p < L$, let $p = p + 1$, go to step 3; if $p = L$, go to step 5;
Step 5: Update the state of the system. Let $t = t + 1$, go to step 2.

We can obtain the real-time speed and position of each train through the TC algorithm, which are made into a train timetable. According to the actual train timetable and the planned train timetable, we can calculate the total delay of trains, the delayed times and so on. The TC algorithm offers the basis of the train routing problem and train scheduling problem. The detailed algorithm to obtain a train schedule can be seen in Appendix A.

4.2. Genetic algorithm based on train control (GATC)

When the size of the railway network is larger, the executive time of accurate algorithm is longer and the time complexity is exponential order. So the accurate algorithm is not appropriate for the large-scale network. In this case, what we need is an efficient algorithm which can get an approximate solution in a finite time. Genetic algorithm can guarantee the finiteness of executive time. In addition, as a heuristic algorithm, genetic algorithm can generally get a better solution with good genetic operators. These genetic operators should be determined through simulation experiments time and again.

Wei and Randy (2006) utilized genetic algorithm to optimize transit route network design. Ji et al. (2009) used a hybrid genetic algorithm for train sequencing problem. Abbas et al. (2011) proposed a genetic-based algorithm for improving train movement. Yang et al. (2012) proposed a new approach to optimize trains movement by using genetic algorithm. Selim and Ismail (2013) used a genetic algorithm to solve train re-scheduling problem on single-track railways. In this paper, we design an improved genetic algorithm called GATC for the train routing problem. The five parts of the proposed GATC are shown as follows.

4.2.1. Chromosome structures and population

The chromosome structure of the proposed GATC in this paper is shown in Fig. 4. Each gene on the chromosome represents a train, which is assigned to a route represented by its allele. The total number of gene on each chromosome represents the total number of trains. Each chromosome represents a feasible solution. Suppose that there are three alternative routes, and then each gene can be assigned to one of the three routes. The single-point crossover operator and random mutation operator are applied to the chromosome structure.

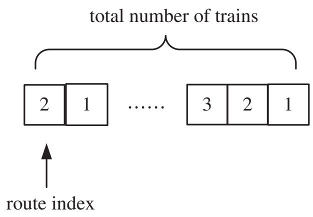

Fig. 4. The chromosome structure.

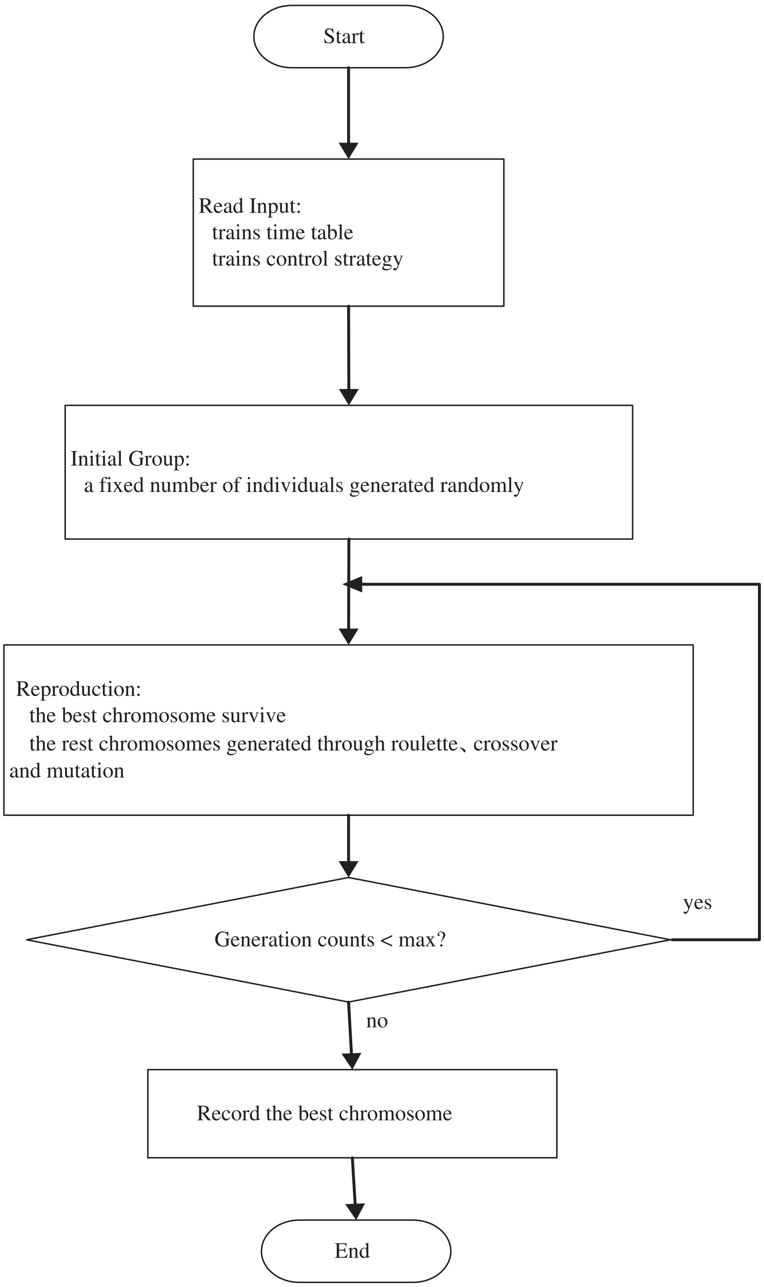

Fig. 5. The generated process of the population.

In our algorithm, the initial population can be generated randomly and each generation maintains a fixed number of the population because that all the randomly generated solutions are feasible solutions. In the evolutionary process, the best individual is picked out and transferred to the next generation; the rest individuals are conducted with the roulette selection, single-point crossover and random mutation process to get the next generation. The population is generated as shown in Fig. 5.

4.2.2. Fitness function

Many goals can be considered in the train routing problem. Three important objective functions are listed in this paper and all of them can express the quality of the solution. If a weighted coefficient is given to each function, we can get a new function. This new function can be used as the fitness function in GATC. The fitness function can be described as formulate (15).

$$F = \alpha \times f_t + \beta \times f_e + \gamma \times f_d \tag{15}$$

The fitness function is restricted by all the constraints in our model and the value of weighted coefficient can be obtained by the sensitivity analysis. The fitness function can judge the quality of the solution. The smaller the fitness function, the better the solution.

4.2.3. Selection process

The GATC uses the roulette to select the new population. In the process of selection, the best individual is saved into the selected new group and the rest individuals are saved or removed according to the roulette rule. The pseudocode of roulette is shown in Fig. 6, cfitness(i) means the accumulated proportion of individual i. Except the best individual, the individual j will remains if a random possibility is between cfitness($j-1$) and cfitness(j), oppositely, the individual j will be removed if there is no random possibility which is between cfitness($j-1$) and cfitness(j). A new population is obtained through the selection.

4.2.4. Crossover procedure

The crossover operator is an important genetic operator, which should be carefully designed. This paper sets $P_c = 0.9$ as the common crossover coefficient and a single-point crossover is used in GATC. Then the population is firstly divided into two equal groups (paternal and maternal group). Two individuals with the same sequence number in the two groups (such as the first individual of the parent group and the first individual of the maternal group) can cross if the random probability is smaller than P_c. If crossover occurs, the system will generate a random integer and the gene whose position equal the

```
begin
for i=1:n // n is the number of individuals
evaluate the fitness and the accumulated proportion of each individual
end
for i=2:n
generate a random number R(i)
  for j=2:n
    if cfitness(j-1)<R(i)<=cfitness(j) //cfitness(i) is the accumulated proportion of individual i
      newpopulation(i)=population(j) // the jth individual in initial population is copied as the ith
                                        individual in new population
    end
  end
 end
end
```

Fig. 6. The code of roulette.

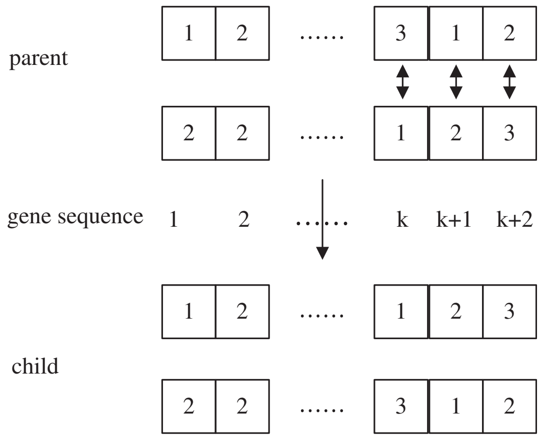

Fig. 7. The process of crossover.

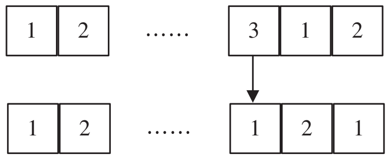

Fig. 8. The process of mutation.

generated number will be selected as the crossover point. The selected gene and its following genes will crossover. After the crossover, the fitness function of the new solution is compared with that of the original solution. If the fitness value is larger than the original fitness value, the crossover is cancelled and the two original individuals retain; otherwise, we adopt the two new individuals. A new population is formed after the crossover.

Supposing that the random number is k, the crossover process is shown in Fig. 7. The gene whose position is k and its following genes are crossed to get two new individuals. Comparing the child with the parent, the two new individuals remain if the fitness value is smaller.

4.2.5. Mutation

This paper uses the standard mutation operator, namely the random mutation. Set the coefficient of mutation $P_m = 0.1$. Randomly generate the mutation point: First, the system randomly generates a route index. Then, the gene whose allele is the route index is chosen to be the mutation point. After the mutation, the fitness function value of the new solution is compared with that of the original solution. If the fitness value is higher than the original fitness value, the mutation is cancelled and the original individual remains; otherwise, we reserve the new mutated individual. The new population is the initial population of the next generation.

For instance, if the random route index generated by system is 3, the mutation process is shown in Fig. 8. The gene whose allele is 3 is chosen to mutate. As the selection and crossover process, if the fitness value is smaller after the mutation, the new individual remains.

Each generation gets a current optimal solution after selection, crossover and mutation. Since the optimal solution of each generation does not participate in the process of genetic process and the better solution is remained in crossover and mutation, the optimal solution of next generation is not worse than the current generation. The convergence of solutions can be guaranteed.

5. Simulation experiments

5.1. A small example

This section takes the above model and algorithms into practice. A one-way double track rail network is shown in Fig. 9, from which we can see that there are three routes on the network. The information of these routes is shown in Table 1. Each route has five nodes. The node A is the origin and node G is the terminal. There are intersections among the three routes. All trains depart from the original station A and arrive at the terminal station G. Trains have to stop at each intermediate station

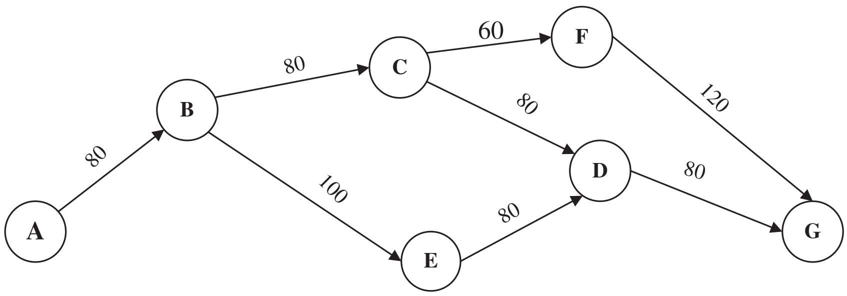

Fig. 9. The high-speed railway network.

Table 1
The information of the three routes.

Route index	The node sequence	Number of nodes	Length of route (km)
Route 1	A-B-C-D-G	5	320
Route 2	A-B-E-D-G	5	340
Route 3	A-B-C-F-G	5	340

Table 2
Properties of the trains.

Train index	Train weight (t)	Maximum speed (km/h)
Train 1	600	240
Train 2	600	300
Train 3	600	300
Train 4	600	240
Train 5	600	300

and the minimum dwelling time is set as 120 s. The capacity of the station is not limited and each station can accommodate numbers of trains.

Assume that the departure time interval at station A is 720 s and there are totally 5 trains including faster trains and slower trains. The maximum speed of faster train is 300 km/h and the maximum speed of slower train is 240 km/h. According to the condition in the real world, the ratio of the mixed trains (faster trains to slower trains) is set as 3:2. In the experiment of this paper, train 1 and train 4 are slower trains and the rest three trains are faster trains. The information of the five trains is shown in Table 2.

A train timetable is given in Table 3 supposing there is no conflict between trains. The timetable illustrates the planned arrival time of each train on each route. By comparing the actual arrival time with the planned arrival time, we can easily find whether one train is delayed in a station.

Utilizing the algorithms proposed in Section 4, an optimal solution of the case above can be found. Here, we use Eq. (15) and set $\alpha = 0.8, \quad \beta = 0.1, \quad \gamma = 0.1$. We set an initial solution that all the trains choose the shortest route (route 1) and easily obtain the initial train schedule as showed in Fig. 10. By utilizing the two algorithms proposed above, we obtain the same optimal result. The optimized train schedule is shown in Fig. 11. In the two figures, the red solid point represents train 1; the blue solid point represents train 2; the black solid point represents train 3; the green solid point represents train 4; the purple solid point represents train 5; the horizontal black solid lines represent the stations.

By contrasting the two figures, we can find that train 1 and train 4 choose route 2 after optimization, which makes the two trains have shorter waiting time at station B. In Fig. 10, train 1 waits a long time to be overtaken by two faster follow-up trains at station B and station C, and train 4 waits at station B to be overtaken by train 5. We can see from Fig. 11 that no train is overtaken after optimization, which decreases the delays of train 1 and train 4 obviously.

The detailed results of initial solution and optimized solution are listed in Table 4. From Table 4 we can find that in initial solution all the trains select route 1, which results in the large train density of route 1. The increase of train density will raise the probability of deceleration and waiting, which will increase the total travel time and delayed times. In addition, the initial objective function value is 4971.5 and the average travel time is 5688 s, while the optimized objective function value is 4815.4 and the average travel time is 5508 s, which is identical calculated by both accurate algorithm and GATC. After optimization, the travel time of train 1 and train 4 decreases by 12.71% and 4.63% respectively, and the travel time of train 3 has a slight increase because the new route is longer than the initial route, while the travel time of train 2 and train 5 have no change because the route of the two faster trains do not change. In summary, the average travel time decreases by 3.16%,

Table 3
The planned train timetable.

Train index	Route index	Arrival time						
		Station A	Station B	Station C	Station E	Station D	Station F	Station G
Train 1	Route 1	7:00	7:23	7:48	–	8:13	–	8:38
	Route 2	7:00	7:23	–	7:53	8:18	–	8:43
	Route 3	7:00	7:23	7:48	–	–	8:08	8:43
Train 2	Route 1	7:12	7:31	7:52	–	8:13	–	8:34
	Route 2	7:12	7:31	–	7:57	8:19	–	8:41
	Route 3	7:12	7:31	7:52	–	–	8:11	8:41
Train 3	Route 1	7:24	7:43	8:04	–	8:25	–	8:46
	Route 2	7:24	7:43	–	8:09	8:31	–	8:53
	Route 3	7:24	7:43	8:04	–	–	8:33	8:53
Train 4	Route 1	7:36	7:59	8:24	–	8:49	–	9:14
	Route 2	7:36	7:59	–	8:29	8:54	–	9:19
	Route 3	7:36	7:59	8:24	–	–	8:44	9:19
Train 5	Route 1	7:48	8:07	8:28	–	8:49	–	9:10
	Route 2	7:48	8:07	–	8:33	8:55	–	9:17
	Route 3	7:48	8:07	8:28	–	–	8:47	9:17

the energy consumption decreases by 2.67% and the delayed times decreases by 100% after optimization. The result demonstrates that accurate algorithm and GATC can get an optimal solution, which can optimize the average travel time, the energy consumption and delayed times simultaneously.

Accurate algorithm can get an optimal solution on a small-scale network, while the executive time will be long whether the size of network or the number of trains is large. GATC can be used to make up for this disadvantage. GATC can get a stable solution or even an optimal solution within a relative short time. This conclusion can be demonstrated by the executive time of the two algorithms in Table 5. In Table 5, when the number of trains is less than 6, the executive times of the two algorithms are approximate. In addition, the accurate algorithm seems more efficient than GATC when the number of train is small. When the number of trains is more than 6, the GATC can still get an optimal solution in a relative short time, while accurate algorithm needs a longer time to get an optimal solution. Therefore, considering the time efficiency, we prefer GATC when the number of trains is more than 6.

5.2. A larger network

In this section, the proposed model and the GATC are applied into a larger high-speed railway network. According to the actual situations in China, the intersections on high-speed railway network are not many. We give a virtual example to test the validity of our method. All rail sections are double-track and we only study the outbound trains. The structure of the high-speed railway network is shown in Fig. 12. For outbound trains, there are three routes in the network. Each route takes A as the starting station and Q as the terminal station, thus every outbound train has the same O–D. Distance between each pair of nodes is given in Figs. 10 and 6 outbound trains will be scheduled into the high-speed railway network in 1 h. The outbound trains are divided into two types: faster trains with maximum speed $V_{\max} = 300$ km/h, slower trains with maximum speed $V_{\max} = 240$ km/h. The trains number ratio is set as 2:1 (i.e. faster trains/slower trains = 2:1) and trains will depart from station A at every 10 min. The weight of each train is also set as 600 t.

The detailed comparison of the accurate algorithm and GATC has been analyzed in Section 5.1. Here, we simply show the results of the two algorithms in this example. The results of accurate algorithm and GATC are shown in Table 6. The total travel time obtained by accurate algorithm and GATC are 1548 min and 1564 min. The energy consumption gained by accurate algorithm and GATC are 92,280 kW and 93,840 kW. Besides, the fitness value calculated by accurate algorithm and GATC are 530.436 and 531.108, which means that the relative error is less than 0.13%. However, the execute time of the accurate algorithm is more than 5 times of GATC. All of those signified that GATC is the better choice when more time efficiency was considered.

6. Sensitivity analysis and perturbation analysis

6.1. Sensitivity analysis

Combined with the experiment in Section 5, we will discuss the influence of the parameter α and β on the objective function value F in Eq. (15) and the deviation f in Eq. (16). The value of F with different α and β is shown as Table 7, here $\gamma = 1 - \alpha - \beta$.

We can get the relationship between α, β and F, shown as in Table 7. From Table 7 we can see that with a fixed γ the value of F is increasing with the increase of α. It indicates that f_t accounts for a larger proportion than f_e. We can also find that the

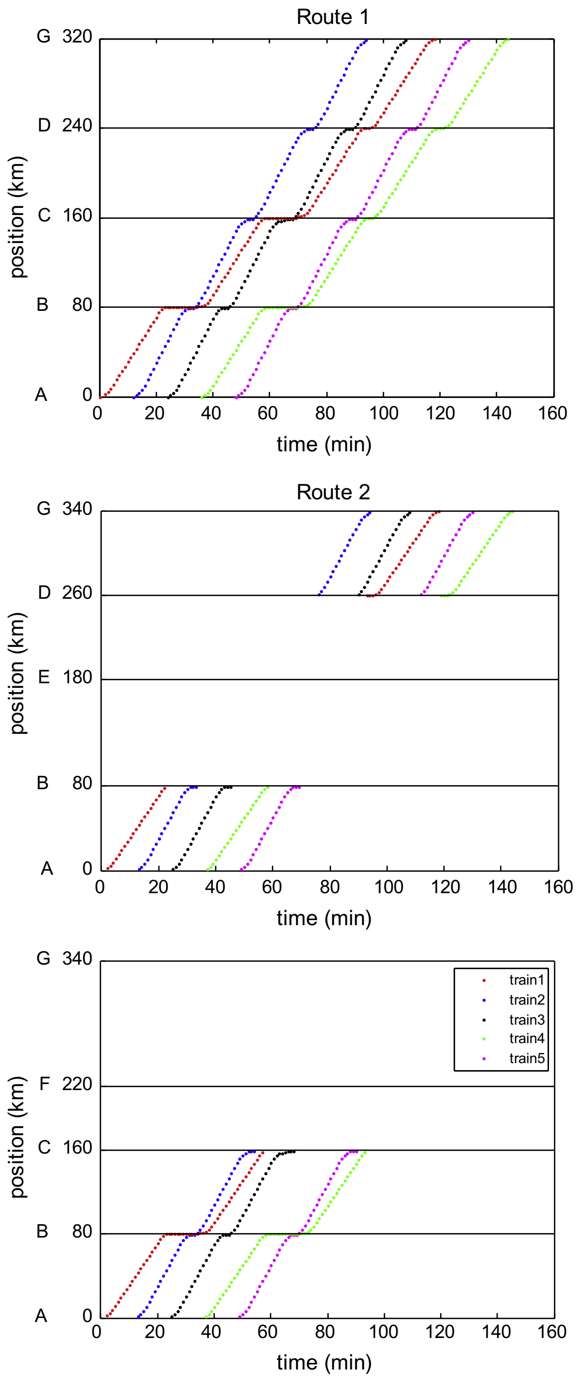

Fig. 10. The initial train schedule.

value of F is decreasing with the increase of γ with a fixed α similarly, which indicates that f_d accounts for a smaller proportion than f_e.

Here, a sensitivity function is defined in this paper to evaluate the degree of deviation:

$$f = \alpha \times |f_t - \bar{f}_t|/\bar{f}_t + \beta \times |f_e - \bar{f}_e|/\bar{f}_e + \gamma \times |f_d - \bar{f}_d|/\bar{f}_d \tag{16}$$

where $\bar{f}_t$ means the expected average travel time of trains, $\bar{f}_e$ means the expected energy consumption and $\bar{f}_d$ means the expected delayed times. $\bar{f}_t$ is set as 5000 s, $\bar{f}_e$ is set as 3000 kW and $\bar{f}_d$ is set as 1. The value of f with different α and β is shown

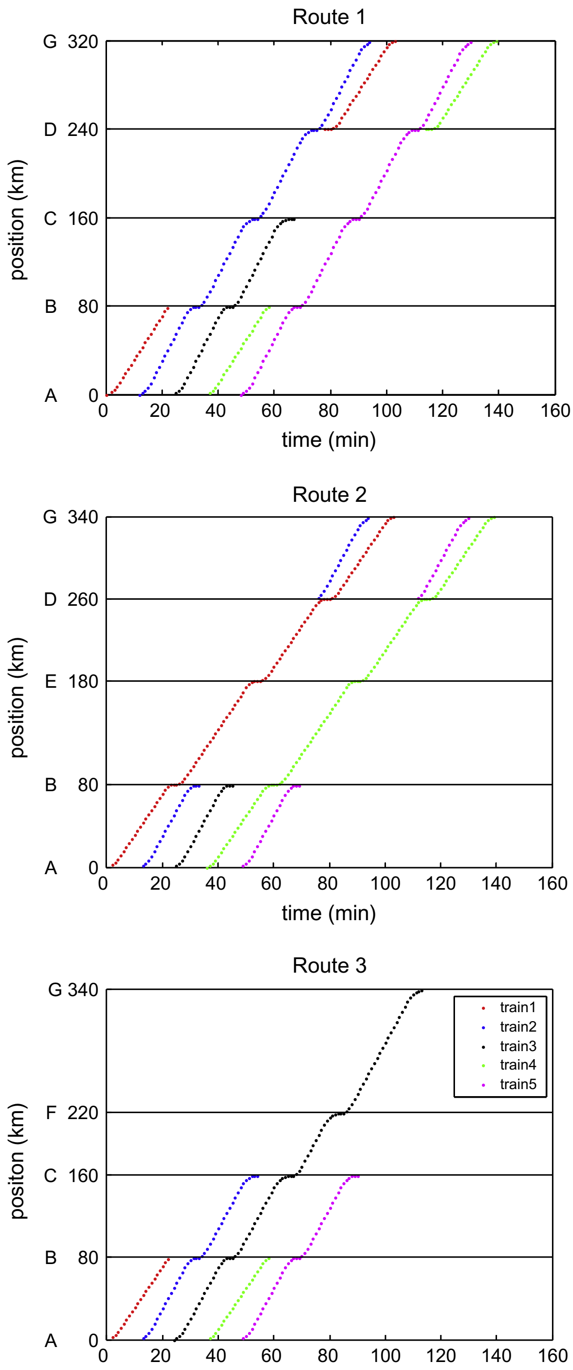

Fig. 11. The optimized train schedule.

in Table 8. In Table 8, with a fixed α, the value of f is decreasing with the increasing of β; with a fixed β, the value of f is decreasing with the increasing of α. In general, f reaches the smallest value with $\alpha = 0.8$, $\beta = 0.1$ and $\gamma = 0.1$. So we utilize $\alpha = 0.8$, $\beta = 0.1$ and $\gamma = 0.1$ in this paper to minimize the degree of deviation.

6.2. Perturbation analysis

In the real world, some unpredictable events may occur in the railway network such as natural disaster and weather. Since these events cannot be controlled by human, trains may travel not in accordance with the planned train schedule absolutely. These unpredictable events are seen as perturbations. The perturbation will have a certain influence on the operation

Table 4
The results of initial solution and optimized solution.

Train index	Initial solution				Accurate algorithm solution/GATC solution			
	Route	Travel time (s)	Energy consumption (kW)	Delayed times	Route	Travel time (s)	Energy consumption (kW)	Delayed times
Train 1	1	7080	4202.4	9	2	6180	4090.4	0
Train 2	1	4920			1	4920		
Train 3	1	5040			3	5340		
Train 4	1	6480			2	6180		
Train 5	1	4920			1	4920		

Table 5
The executive time of two algorithms.

Train number	Accurate algorithm Executive time (s)	GATC Executive time (s)
3	6.7723	12.1575
4	16.9479	30.2672
5	51.2168	98.2592
6	512.0844	127.3259
7	–	259.7557
8	–	600.2992

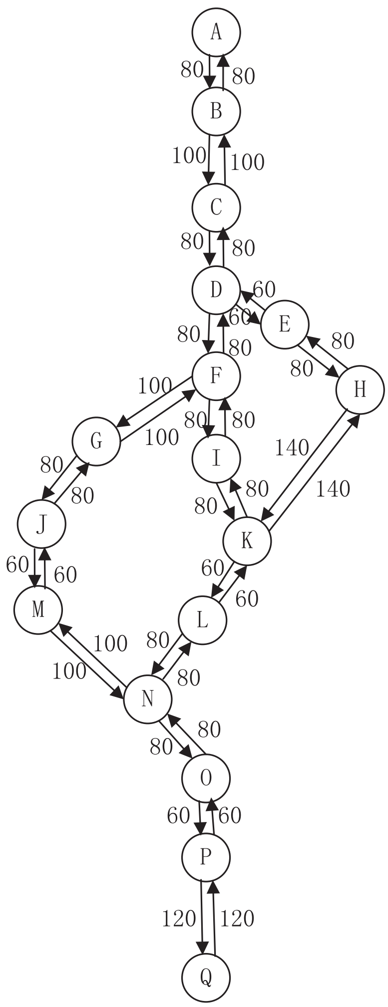

Fig. 12. The larger high-speed railway network.

Table 6
The results of accurate algorithm and GATC.

Train index	Accurate algorithm solution				GATC solution			
	Route	Travel time (min)	Energy consumption (kW)	Execute time (min)	Route	Travel time (min)	Energy consumption (kW)	Execute time (min)
Train 1	1	244	92,880	86.48534	1	244	93,840	16.40575
Train 2	1	244			1	244		
Train 3	3	288			2	288		
Train 4	1	249			3	257		
Train 5	1	245			3	253		
Train 6	1	278			1	278		

Table 7
The value of F with different α and β.

F	$\alpha = 0.1$	$\alpha = 0.2$	$\alpha = 0.3$	$\alpha = 0.4$	$\alpha = 0.5$	$\alpha = 0.6$	$\alpha = 0.7$	$\alpha = 0.8$
$\beta = 0.1$	959.8	1510.6	2061.4	2612.2	3163.0	3713.8	4264.6	4815.4
$\beta = 0.2$	1368.9	1919.7	2470.5	3021.3	3572.1	4122.9	4673.7	–
$\beta = 0.3$	1777.9	2328.7	2879.5	3430.3	3981.1	4531.9	–	–
$\beta = 0.4$	2187.0	2737.8	3288.6	3839.4	4390.2	–	–	–
$\beta = 0.5$	2596.0	3146.8	3697.6	4248.4	–	–	–	–
$\beta = 0.6$	3005.0	3555.8	4106.6	–	–	–	–	–
$\beta = 0.7$	3414.1	3964.9	–	–	–	–	–	–
$\beta = 0.8$	3823.1	–	–	–	–	–	–	–

Table 8
The value of f with different α and β.

f	$\alpha = 0.1$	$\alpha = 0.2$	$\alpha = 0.3$	$\alpha = 0.4$	$\alpha = 0.5$	$\alpha = 0.6$	$\alpha = 0.7$	$\alpha = 0.8$
$\beta = 0.1$	0.8270	0.7372	0.6474	0.5575	0.4677	0.3778	0.2880	0.1982
$\beta = 0.2$	0.7439	0.6541	0.5642	0.4744	0.3845	0.2947	0.2049	–
$\beta = 0.3$	0.6608	0.5709	0.4811	0.3913	0.3014	0.2116	–	–
$\beta = 0.4$	0.5776	0.4878	0.3980	0.3081	0.2183	–	–	–
$\beta = 0.5$	0.4945	0.4047	0.3148	0.2250	–	–	–	–
$\beta = 0.6$	0.4114	0.3215	0.2317	–	–	–	–	–
$\beta = 0.7$	0.3282	0.2384	–	–	–	–	–	–
$\beta = 0.8$	0.2451	–	–	–	–	–	–	–

of trains. A perturbation analysis will provide the effective information for rerouting and rescheduling. Two cases of perturbations are considered and analyzed as follows.

Case 1: Train 1 encounters with an emergency event at the 660th seconds and has to brake immediately. The time of troubleshooting is 600 s and train 1 has to wait for 600 s.

The schedule after perturbation of case 1 is shown as Fig. 13.

6.2.1. Delay analysis

Before perturbation, the average travel time of trains is 5508 s, the energy consumption is 4090.4 kW and the delayed times is 0. After perturbation, the average travel time of trains is 6552 s, the energy consumption is 5989.2 kW and the delayed times is 20. The average travel time of the trains increases by 18.95% and the energy consumption increases by 46.42% after perturbation. Compared with the schedule before perturbation, it can be represented that the delay phenomenon is obvious in section (A, B), and the delays would also propagate on the following trains. In addition, train 2 and train 3 are greatly affected by the propagation of delays. To keep the safe distance, frequent deceleration and acceleration are manipulated, which lead to the increase of energy consumption. Besides, the delays occur before the first station and then it propagate on the later stations, which indicate that the delayed times will increase after perturbation.

6.2.2. Speed analysis

The trains speed before and after perturbation are compared in Table 9, which indicate that the delay of the head train would affect the follow-up trains. The average speed of all trains, the average speed of faster trains and the

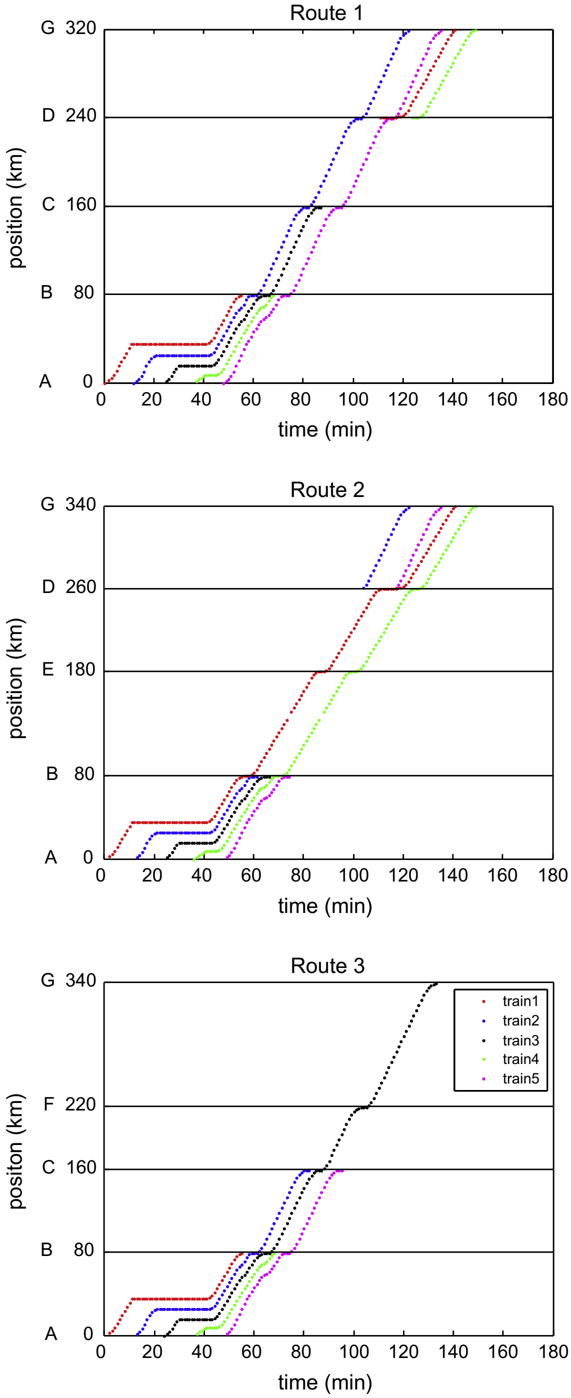

Fig. 13. The schedule of trains after perturbation of case 1.

average speed of slower trains decreases by 14.84%, 14.41% and 15.59% respectively. Although train 2 and train 3 are the faster trains, train 1 would not be overtaken by these two trains. Train 2 and train 3 have to stop behind train 1 to guarantee the safety, thus the average speed of the faster trains decreases. Except that train 1 is delayed on section (A, B), train 4 is also a slower train and affected by train 3 on section (A, B), which lead to the decrease of the average speed of the slower trains as well.

Table 9
The speed comparison before and after perturbation in case 1.

	Speed of trains (km/h)		
	Average speed of all trains	Average speed of faster trains	Average speed of slower trains
Before perturbation	218.7245	232.5020	198.0582
After perturbation	186.2726	199.0011	167.1798
Decrease rate (%)	14.84	14.41	15.59

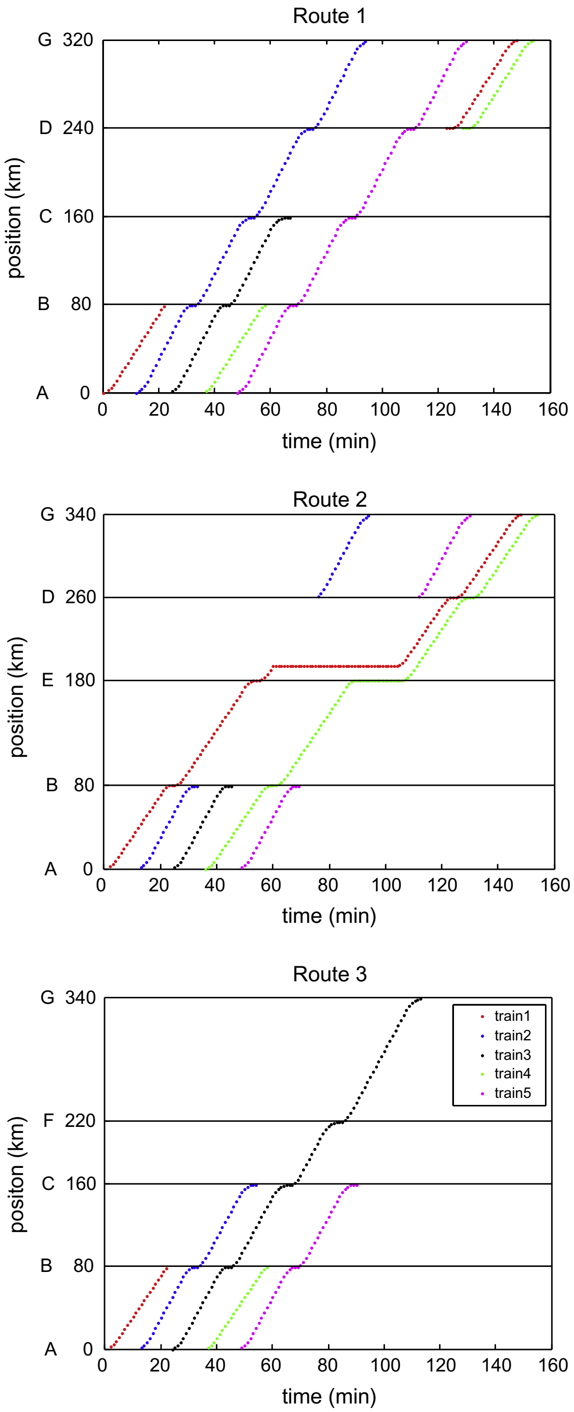

Fig. 14. The schedule of trains after perturbation of case 2.

Table 10
The speed comparison before and after perturbation in case 2.

	Speed of trains (km/h)		
	Average speed of all trains	Average speed of faster trains	Average speed of slower trains
Before perturbation	218.7245	232.5020	198.0582
After perturbation	201.6450	232.5020	155.3596
Decrease rate (%)	7.81	0	21.59

6.2.3. Capacity analysis

The delays caused by the head train not only decrease the average speed of trains but also decrease the traffic capacity. The traffic capacity here is defined as the total number of trains that arrive at the terminal station in 1 day. The traffic capacity decreases from 51 to 49 after perturbation, thus the traffic capacity decreases by 3.92%.

Case 2: Due to the malfunction, section (E, D) in the rail network is not available from the 3600th seconds to the 7200th seconds.

The schedule after perturbation is shown as Fig. 14.

6.2.4. Delay analysis

Before perturbation, the average travel time of trains is 5508 s, the energy consumption is 4090.4 kW and the delayed times is 0. After perturbation, the average travel time of trains is 6228 s, the energy consumption is 4240.4 kW and the delayed times is 5. Thus the average travel time of the trains increases by 13.07% and the energy consumption increases by 3.67% after perturbation. From Fig. 14 we can find that the delays of trains, especially the delays of train 1 and train 4, are obvious in section (E, D) compared with the schedule before perturbation. Section (E, D) has not been forbidden when train 1 travels into section (E, D), so train 1 stops on section (E, D) for 3600 s. While train 4 arrives at station E before the ban is removed, thus train 4 has to wait at station E until the section (E, D) opens. The emergency braking and acceleration of train 1 on section (E, D) increase the energy consumption slightly. In addition, the delays of train 1 and train 4 on section (E, D) increase the delayed times.

6.2.5. Speed analysis

Table 10 has listed the speed comparison before and after perturbation. From Table 10 we can find that the ban of the section (E, D) affects train 1 and train 4. The average speed of all trains decreases by 7.81% and the average speed of slower trains decreases by 21.59%, while the average speed of faster trains has not changed. The section (E, D) is located at route 2 and chosen by only two trains (train 1 and train 4). Delays on section (E, D) will increase the travel time of train 1 and train 4, which will lead to the decrease of the average speed of slower trains. The three faster trains select route 1 or route 3 respectively, so the three faster trains will not be affected by the ban of section (E, D). Thus the average speed of the slower trains decreases, while the average speed of faster trains has no change.

6.2.6. Capacity analysis

The delays caused by the ban of section (E, D) do not only decrease the average speed of trains but also decrease the traffic capacity. The traffic capacity changes from 51 to 46 after perturbation. It means that the traffic capacity decreases by 9.80%.

7. Conclusions

This paper has proposed an efficient method to solve the optimization of train routing problem on a one-way double track rail network. This method can be used to get a practicable train schedule quickly and provide an important reference for the train plan. First of all, considering the three targets (i.e. minimize the average travel time of trains, minimize the energy consumption and minimize the delayed times), a multi-objective optimization model for train routing problem is constructed based on train control strategies. Secondly, two detailed algorithms for train routing problem are designed, which are called accurate algorithm and GATC respectively. accurate algorithm is a quick and accurate algorithm to solve the train routing problem when the size of network is small, while GATC is appropriate for the train routing problem on a larger-scale network. In addition, a sensitivity analysis is made to probe the influence of coefficients on the objective function F and the deviation degree f. The ideal coefficients are found to minimize the deviation degree f in our model. Finally, two cases of disturbance are analyzed. The analysis results show the influences of perturbations on train delay, train speed and traffic capacity, which can provide valuable information for rerouting and rescheduling.

Acknowledgements

The research is supported in part by the National Basic Research Program of China (Grant No. 2012CB725400), the National Natural Science Foundation of China (Grant No. 71131001-1), the Research Foundation of State Key Laboratory of Rail Traffic Control and Safety (Nos. RCS2012ZZ001 and RCS2012ZT001), Beijing Jiaotong University.

Appendix A

Based on train control strategies, the value of the fitness is calculated according to the simulation algorithm. Moreover, a train schedule can be obtained by the simulation algorithm. The detailed simulation algorithm is presented as follow:

Notations

g	the distance between ith train and its front station
c	the distance between ith train and its front train
N	the terminal time of the simulation
D	the position of the front station
F_i	the tractive power of ith train
L_i	the length of the route which is chosen by ith train
R_i	the route which is assigned to ith train
X_c	the braking distance of ith train
w_i	the delay times of ith train
$f_{\min}$	the minimum protective distance between two adjacent trains
$A_{i,a}$	the actual arrival time of ith train
$P_{i,m}$	$P_{i,m} = 1$ if ith train is prior than mth train; $P_{i,m} = 0$ if mth train is prior than ith train
$V_{p,i}(t)$	at simulation time t, the speed of ith train at position p
$h_i(1,p)$	if ith train is at position p, $h_i(1,p) = 1$; else $h_i(1,p) = 0$
$\overline{V}_{p,i}(t+1)$	the temporary speed of ith train at time $t+1$

Algorithm.

Step 1: (Initialization) set the time $t = 0$, $F_i = 0$, $b = 0$.
Step 2: (Initialization) set the position $p = 0$, the speed of ith train $V_{p,i}(0) = 0$. $t = 1, 2, \ldots, N$.
Step 3: If $h_i(1,p) = 1$ (i.e. ith train is at position p), go to step 4; otherwise, go to step 6.
Step 4: If $p = L_i$, record $A_{i,a} = t$. Set $k = 1,\ g = D - p$.
Step 4.1: If $p + k \leqslant L_i,\quad h_m(1, p+k) = 1$ *and* $R_m = R_i$ (i.e. ith train and mth train select the same route) or $p + k > L_i$, let $c = \min(k, g)$, go to step 5, here $m \in \{1, 2, \ldots, i-1, i+1, \ldots, N\}$, L_i is the length of the chosen route; otherwise, go to step 4.2.
Step 4.2: If $p + k \leqslant L_i,\ h_m(1, p+k) \neq 1$, let $k = k + 1$, go to step 4.1.
Step 5: Determine the speed and location of the focal ith train.
Step 5.1: If ith train is a faster train.
Case 1: If $p \neq D$ and $c < g$.
(a) If $c \leqslant \max(X_c + f_{\min}, f_{\min})$, let $\overline{V}_{p,i}(t+1) = \max(0, V_{p,i}(t) - a)$,
then $V_{p,i}(t+1) = \min(\overline{V}_{p,i}(t+1), \sqrt{2a(c - f_{\min}) + V_{p+k,m}(t)^2})$.
here $X_c = (V_{p,i}(t)^2 - V_{p+k,m}(t)^2)/2a$;
(b) If $c > \max(X_c + f_{\min}, f_{\min})$, let $V_{p,i}(t+1) = \min(V_{\max 2}, V_{p,i}(t) + a)$.
Case 2: If $p \neq D$ and $c = g$.
(a) If $g \leqslant V_{p,i}(t)^2/2a$, let $V_{p,i}(t+1) = \min(\overline{V}_{p,i}(t+1), \sqrt{2ag})$,
where $\overline{V}_{p,i}(t+1) = \max(0, V_{p,i}(t) - a)$;
(b) If $g > V_{p,i}(t)^2/2a$, let $V_{p,i}(t+1) = \min(V_{\max 2}, V_{p,i}(t) + a)$.
Case 3: If $p = D$.
(a) If $b = 0 \quad and \quad A_{i,k,a} > A_{i,k,o}$, then $w_i = w_i + 1$. If $b \leqslant e$, let $V_{p,i}(t+1) = 0, \quad b = b + 1$;
(b) If $b > e$ and $c \leqslant \max(X_c + f_{\min}, f_{\min})$, let $V_{p,i}(t+1) = 0, \quad b = b + 1$;
(c) If $b > e$ and $c > \max(X_c + f_{\min}, f_{\min})$, let $V_{p,i}(t+1) = \min(V_{\max 2}, V_{p,i}(t) + a),\ b = 0$.
Then if $V_{p,i}(t+1) > V_{p,i}(t)$, calculate the energy consumption f_i, let
$F_i = F_i + f_i$. Go to step 5.3.
Step 5.2: If ith train is a slower train.
Case 1: If $p \neq D$ and $c < g$.
(a) If $c \leqslant \max(X_c + f_{\min}, f_{\min})$, let $\overline{V}_{p,i}(t+1) = \max(0, V_{p,i}(t) - a)$, then let $V_{p,i}(t+1) = \min(\overline{V}_{p,i}(t+1),$
$\sqrt{2a(c - f_{\min}) + V_{p+k,m}(t)^2})$;
(b) If $c > \max(X_c + f_{\min}, f_{\min})$, let $V_{p,i}(t+1) = \min(V_{\max 1}, V_{p,i}(t) + a)$.

(*continued on next page*)

Case 2: If $p \neq D$ and $c = g$.
(a) If $g \leqslant V_{p,i}(t)^2/2a$, let $V_{p,i}(t+1) = \min(\overline{V}_{p,i}(t+1), \sqrt{2ag})$, where $\overline{V}_{p,i}(t+1) = \max(0, V_{p,i}(t) - a)$;
(b) If $g > V_{p,i}(t)^2/2a$, let $V_{p,i}(t+1) = \min(V_{\max 1}, V_{p,i}(t) + a)$.
Case 3: If $p = D$.
(a) If $b = 0 \quad and \quad A_{i,k,a} > A_{i,k,o}$, then $w_i = w_i + 1$. If $b \leqslant e$, $V_{p,i}(t+1) = 0, \quad b = b+1$;
(b) If $b > e$ and $c \leqslant \max(X_c + f_{\min}, f_{\min})$, let $V_{p,i}(t+1) = 0, \quad b = b+1$;
(c) If $b > e$, $c > \max(X_c + f_{\min}, f_{\min})$ and $P_{i,m} = 0$, where $h_m(1, p-j) = 1$, then let $V_{p,i}(t+1) = 0, \quad b = b+1$;
(d) If $b > e, \ c > \max(X_c + f_{\min}, f_{\min})$ and $P_{i,m} = 1$, where $h_m(1, p-j) = 1$, then let $V_{p,i}(t+1) = \min(V_{\max 1}, V_{p,i}(t) + a), \quad b = 0$.
Then if $V_{p,i}(t+1) > V_{p,i}(t)$, calculate the energy consumption f_i, let $F_i = F_i + f_i$, go to step 5.3.
Step 5.3: Let $\overline{h}_i(1, p + V_p(t+1)) = 1$,
$\overline{V}_{p+v_p(t+1),i}(t+1) = V_{p,i}(t+1)$. Go to step 6.
Step 6: If $p < L_i$, let $p = p + 1$, go to step 3; otherwise, go to step 7;
Step 7: Update the state of the system:
$h_i(1,p) = 0, h_i(1, p + V_p(t+1)) = \overline{h}_i(1, p + V_p(t+1))$; $V_{p,i}(t+1) = 0, \quad V_{p+v_p(t+1),i}(t+1) = \overline{V}_{p+v_p(t+1),i}(t+1)$.
Then, let $t = t + 1$. if $t < T$, go to step 2; else if $t = T$, algorithm is end, the objective function
$F = 0.8 \times \sum_{i=1}^{N}(A_{i,a} - A_{i,o})/N + 0.1 \times \sum_{i=1}^{N} F_i + 0.1 \times \sum_{i=1}^{N} w_i$.

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